

Determining critical phase transition temperature in an Ising model simulation

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Room for abstract.

I. INTRODUCTION

A. The Ising model

The Ising model named after the German physicist Erns Ising is a model in statistical mechanics. The model consists of discrete magnetic spin dipole moments that take values ± 1 ("up" and "down" respectively) arranged on a lattice. For the purposes of our simulation we used a square lattice. The spins interact with neighbouring spins. The model exhibits phase transitions that can be associated with the gas-liquid transition or spontaneous magnetisation.

B. Aim of study

In this article we look at determining at what temperature does the Ising model undergo a phase transition. It is known from theory that under a certain critical temperature the model stabilises into completely up or down state. Conversely, above this critical, on average, the model remains a random mixture of up and down spins.

II. THEORY

The theoretical value of the critical temperature was predicted by Lars Onsager[1] to be

$$T_c = \frac{2}{\ln(1 + \sqrt{2})}. \quad (1)$$

The interaction energy between two spins is given by

$$E_{\text{int}} = -J s_i s_j, \quad (2)$$

where $J > 0$ is the coupling constant for ferromagnetic interactions and the s_i being the value of spin of spin i . We define the reduced interaction energy as $E_r = \frac{E_{\text{int}}}{J} = -s_i s_j$. The reduced interaction energy with the j nearest neighbours is then

$$E_r = \sum_{\langle j \rangle} s_i s_j, \quad (3)$$

III. METHODOLOGY

A. Metropolis algorithm

To simulate the evolution of the Ising model we use a Monte Carlo simulation using the Metropolis algorithm. In our implementation we consider the von Neumann neighbourhood of only 4 neighbouring spins in the $\pm x$ and $\pm y$ directions. We also use a periodic boundary condition such that if we denote by $n \times n$ the size of the lattice, we have that the spin at position (n, i) is a neighbor of spin at position $(1, i)$ and in the same way also spin at i, n is neighbouring the spin at $(i, 1)$. At each iteration of the algorithm a random position is chosen in the square lattice of spins. The energy needed to flip the spin from the up to the down state or vice versa is then calculated from Eq. (3) as

$$\Delta E = E_{\text{before}} - E_{\text{after}} = \sum_{\langle j \rangle} (-s_i) s_j - \sum_{\langle j \rangle} s_i s_j = -2 \sum_{\langle j \rangle} s_i s_j. \quad (4)$$

We then have 2 cases. If $\Delta E \leq 0$, we flip the spin orientation. If $\Delta E > 0$, we accept the flip with probability $p = e^{\frac{\Delta E}{T_r}}$. In order to do this, we sample a random variable $X \sim \text{Unif}(0, 1)$ and check $X < p$, in which case we accept the flip and leave the spin unflipped otherwise.

We know that for values of the reduced temperature below the critical temperature i.e. $T_r < T_c$ the system gets to an equilibrium state where the absolute value of the average magnetisation goes to 1[2].

B. Determining parameters for final simulation

TODO: <Explain how we arrived at simulation parameters>

IV. RESULTS

TODO: <Give results>
TODO: <Give figures of results>

V. COMPARISON TO THEORY

A. Theoretical solution for the critical phase transition temperature

TODO: <Give derivation of critical temperature in the Ising model>

B. Comparison of theory and our results

TODO: <Compare our results with theoretical values>

VI. CONCLUSIONS

Conclusion.

[1] Lars Onsager. Crystal statistics. i. a two-dimensional model with an order-disorder transition. *Physical Review*, 65(3-4):117-149, 1944.

[2] Stephen J. Blundell and Katherine M. Blundell. *Concepts in Thermal Physics*. Oxford University Press, Oxford, 2nd edition, 2010.